

HW 6 Math 638 Differential Geometry II, Winter 2026

This homework is due on Week 7, Wednesday, Feb 18, 2026.

Q6.1. ([Lee, 2nd, 12-5, p. 325]) Let V be a n -dimensional real vector space. Show that

$$\dim_{\mathbb{R}} \Sigma^k(V^*) = \frac{(n+k-1)!}{k!(n-1)!} = \binom{n+k-1}{k},$$

where $\Sigma^k(V)$ is the vector space of symmetric k -tensor.

Q6.2. Show that any symmetric bilinear map $b : V \times V \rightarrow W$ can be recovered from the associated quadratic form $q(v) \doteq b(v, v)$.

(Hint 1: Consider the following case. A symmetric bilinear map on $\mathbb{R}^2$,

$$B((x^1, x^2), (y^1, y^2)) \doteq 3x^1y^1 + \frac{3}{2}x^1y^2 + \frac{3}{2}x^2y^1 - x^2y^2,$$

we can recover it from the the associated quadratic form

$$Q(x, y) \doteq B((x, y), (x, y)) = 3x^2 + 3xy - y^2$$

by the following formula

$$B((x^1, x^2), (y^1, y^2)) = \frac{1}{2!} \left. \frac{\partial^2}{\partial s_1 \partial s_2} \right|_{s_1=s_2=0} Q(s_1(x^1, x^2) + s_2(y^1, y^2))$$

using calculus.

Hint 2: When $V = W = \mathbb{R}$ we can rewrite $b(x, y) = xy = \frac{1}{2}(q(x+y) - q(x) - q(y))$.

Q6.3. (3a) Show that any covariant 2 tensor can be written as a sum of a symmetric 2 tensor and an alternating 2 tensor.

(3b) ([Lee, 2nd, 12-7, p. 326]) Give a covariant 3 tensor who can not be written as a sum of a symmetric 3 tensor and an alternating 3 tensor. Give some reason.

Q6.4. Let V be a n -dimensional real vector space. Let T_1, T_2, T_3 be three covariant tensor and let ω be a covector. Show that $\text{Alt}(T_1 \otimes \omega \otimes T_2 \otimes \omega \otimes T_3) = 0$.

Q6.5. Let V be a n -dimensional real vector space. Let $\omega_1, \dots, \omega_k$ be k vectors in the dual space V^* . Show that $\text{Alt}(\omega_1 \otimes \dots \otimes \omega_k) = 0$ when $\omega_1, \dots, \omega_k$ are linearly dependent.

Q6.6. ([Lee, 2nd, 12-1, p. 324]) Give an example of finite-dimensional vector spaces V and W and a specific element $\alpha \in V \otimes W$ that cannot be expressed as $v \otimes w$ for some $v \in V$ and $w \in W$.